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An Optimized ALT Pipeline

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Initial Literature Review

Recent advances in deep learning, particularly the adoption of weakly-supervised transformer architectures trained on large-scale corpora, have produced automatic speech recognition (ASR) models capable of reliably transcribing speech [1]. Extending these capabilities to singing is valuable, as robust automatic lyric transcription (ALT) enables a number of useful downstream applications, subtitling and lyrics-based retrieval [1, 2, 3]. Recognizing singing, however, presents a number of domain-unique problems: pronunciation and diction vary across genres and individual singers, singing can exhibit more pitch variation than typical speech, and songs often contain high levels of interfering noise due both to background vocals and instrumental accompaniment [2, 4, 5]. The training of ASR systems for singing is further hampered by the a lack of large, high-quality, singing-specific datasets [2, 6].

Despite the aforementioned differences between normal speech and singing, robust ASR models trained on large amounts of speech, such as Whisper [1], generalize fairly well to singing [2, 5, 6]. As a result, their use as zero-shot foundation models for automatic lyrics transcription (ALT) has become increasingly common, underpinning the current state-of-the-art system, LyricWhiz [6]. LyricWhiz pairs Whisper as the "ear" with GPT-4 as the "brain" behind its transcriptions [6]: Whisper produces draft transcriptions from raw audio, which GPT-4 then selects among and corrects to produce a final prediction. LyricWhiz is evaluated on the Jamendo [7], Hansen [8], and DSing [9] datasets, achieving word error rates (WERs) of 24.25%, 7.85%, and 12.99%, respectively [6].

LyricWhiz performs well on both short-form audio clips and full-length songs. The latter use case does, however, introduce an additional challenge: Whisper processes audio in fixed windows of at most 30 seconds [1], so a full song must first be divided into segments short enough for the model to consume. Syed et al. [5] address this challenge directly, introducing RMS-VAD: an algorithm, building on prior work [10], that leverages the amplitude of source-separated audio as for voice activity detection. RMS-VAD first computes the root mean square (RMS) amplitude of the source-separated vocal signal to identify segment boundaries, then applies the cut-and-merge strategy leveraged by WhisperX [11] to consolidate these into segments of appropriate length for transcription.

Syed et al. evaluate their approach on Jam-ALT [12], a benchmark of manually corrected

lyric transcriptions for full-length songs, designed to address inconsistencies in earlier lyric transcription datasets regarding punctuation, casing, and non-lexical vocables. With this improved segmentation strategy, lyric transcriptions produced by Whisper alone approach the accuracy of those produced by LyricWhiz without leveraging LLM post-processing, achieving a WER of 24.73% compared to LyricWhiz's 23.70% on the Jam-ALT dataset [5].

Though source separation is integral to the segmentation algorithm proposed by Syed et al. [5], it is not used to directly improve the audio fed to Whisper for transcription; this design choice is consistent with findings that separation artifacts can cause Whisper to hallucinate [5, 12]. Source separation is, additionally, computationally expensive, with Demucs requiring upwards of 40 million parameters [13]. The authors themselves note that identifying alternative segmentation algorithms represents a valuable line of inquiry [5], and several such approaches that employ comparatively lightweight neural architectures have shown promise in the literature [14, 15].

Although Syed et al. [5] substantially improve long-form segmentation, their pipeline does not reintroduce LLM-based post-processing. This omission is notable given that LyricWhiz's own ablation study attributes some of its own performance gain to GPT-based ensembling and correction [6] (see table 3 of the original paper), and that generative error correction (GER) has independently been shown to substantially reduce WER in conventional, non-singing ASR [16, 17, 18]. Reintroducing such post-processing into Syed et al.'s improved segmentation pipeline therefore represents a promising, currently unexplored direction for further reducing WER in long-form lyric transcription.

Taken together, this literature reveals two under-explored directions for improving long-form ALT. While RMS-VAD improves upon Whisper's native segmentation, it remains dependent on computationally expensive source separation [13]; prior work on lightweight, feature-based singing voice detection suggests that reasonably accurate segmentation can be achieved with substantially less computational overhead [14, 15]. Separately, although Generative Error Correction (GER) has proven effective for conventional ASR [16, 17, 18] and underlies a significant amount LyricWhiz's own performance [6], it has not been systematically explored on sung speech, nor in combination with improved segmentation for long-form ALT specifically. To the best of my knowledge, no existing work has investigated the joint effect of these two directions. This proposal is accordingly guided by the following research questions:

- **RQ1:** Can a less computationally intensive segmentation algorithm achieve comparable or superior performance to source-separation-based VAD for long-form ALT?
- **RQ2:** To what extent can GER reduce WER in this domain?
- **RQ3:** When combined, do improved segmentation and GER yield synergistic improvements in WER, or do the two methods interact negatively?

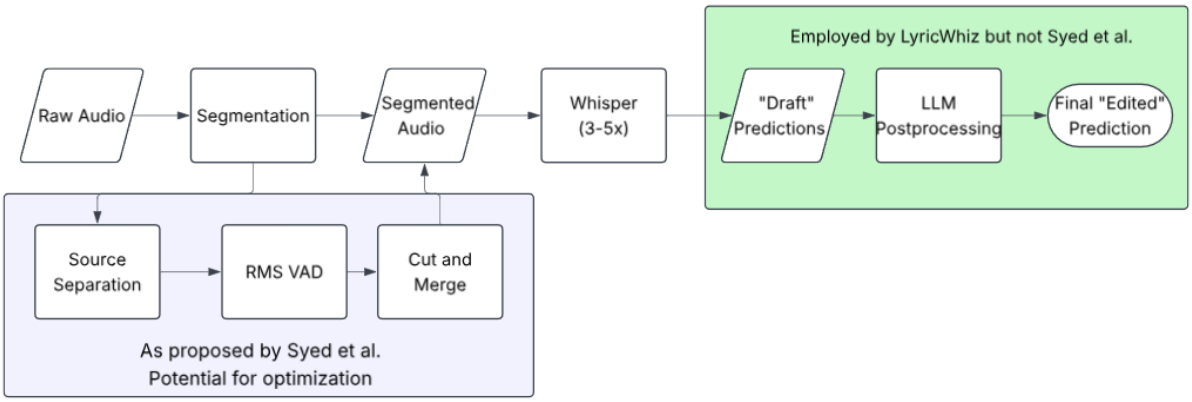


Figure 1: An Automatic Lyric Transcription (ALT) Pipeline, highlighting recent work in the field [5, 6]

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Appendix A: Relevant Results

Method	Jamendo	Hansen	DSing
TDNN-F [8]	76.37	77.59	19.60
CTDNN-SA [44]	66.96	78.53	14.96
Genre-informed AM [12]	50.64	39.00	56.90
MSTRE-Net [13]	34.94	36.78	15.38
DE2-segmented [45]	44.52	49.92	-
W2V2-ALT [22]	33.13	18.71	**12.99**
LyricWhiz (Ours)	**24.25**	**7.85**	13.78
w/o ChatGPT Ens.	28.18	8.07	15.22
w/o Whis. Prompt	33.21	8.75	13.40

Figure 2: Table 3 from the paper that introduced LyricWhiz [6]. LLM postprocessing reduces WER across all assessed datasets.